Separating predictive information from delayed adaptation gains in time-series foundation models

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Abstract

Comparisons of online calibration can confound predictive information, feedback state and retained score history. We hold neural forecasts and reveal schedules fixed to separate these effects. Native-band conformalized quantile regression (CQR) improves interval score by 9.98% over matched, validation-tuned symmetric residual calibration across 124 model–series pairs. A protocol-locked weather/air-quality study shows that rate controls alone can leave a misleading state comparison. On weather at horizon 24, subsequent positive-rate analysis reduces the issued/shared CQR score gap from 6.080% at a fixed window to 0.052% with window selection, before admitting zero. Many selected rates remain very small. At one step, adaptive CQR improves mean score by only 0.36% but reduces local coverage error from 3.15 to 1.43 percentage points; three of four weather pairs improve in every test block. Common-support and asymmetric-residual controls leave a 9.56% positive-output photovoltaic score benefit, with only 86.45% coverage at a 90% target. Full configured AcMCP-CSS, common-history comparisons and exact clipping accounting support the evaluation. The findings distinguish predictive information, configuration-dependent adaptation gains and unresolved reliability gaps.

Keywords: Time-series foundation models, Conformal calibration, Delayed feedback, Probabilistic forecasting, Conditional reliability

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1. Introduction

Frozen time-series foundation models provide point predictions and quantiles without training a new forecasting model for every deployment stream [1, 2, 3, 4]. Their uncertainty estimates can still need calibration. A comparison of complete calibration pipelines, however, combines several choices: which predictive information enters the score, how delayed feedback updates its level, and how much history is retained. A better interval score does not identify which choice supplied the benefit. Nor does it establish reliable coverage in every time period or operating regime.

This paper studies three questions on fixed neural forecasts. First, how much value does native predictive information retain after stronger residual and interval-construction controls? Second, what does feedback adaptation add after learning rate and memory are selected together? Third, which reliability deficits remain when interval scores improve? We use existing calibration methods as controlled components, keeping targets, forecast origins and reveal schedules aligned across each comparison.

The main finding is an interaction between score construction, feedback state and calibration memory. On a protocol-locked weather panel, the issued/shared CQR score gap remains 6.08% after admitting zero and very small learning rates at a fixed window, but falls to about 0.05% with window selection. A subsequent positive-rate analysis produces essentially the same attenuation before zero is admitted (Figure 1). Thus the observed collapse cannot be explained solely by selecting an exactly zero rate. Many selected positive rates are nevertheless tiny: the result concerns how comparisons change near the rolling limit, rather than a proof that substantive feedback mechanisms are equivalent.

A second finding separates average-score and local-reliability benefits. At one step on weather, adaptive CQR has only about 0.36% mean paired score improvement over selected-window Rolling CQR, but mean local coverage error falls from 3.15 to 1.43 percentage points. Shared and issued states coincide at one step, so this is a feedback-level benefit rather than a state-architecture benefit. All three time blocks retain a local improvement for three of the four weather pairs; the fourth selects the same rolling method. This supplies a concrete boundary to an interpretation that feedback has little value merely because its long-horizon score benefit is small.

Native-band information also survives stronger construction controls. On the original PV panel, common nonnegative projection and an asymmetric signed-residual baseline leave a 9.56% positive-output score benefit. Yet CQR coverage on that subset is 86.45% at a 90% target, and a high-

forecast group observable at issue time also under-covers. The resulting contribution is a set of conditional empirical findings about attribution and reliability, supported by exact coverage accounting, new-source confirmation and a forecast-based reproduction artifact.

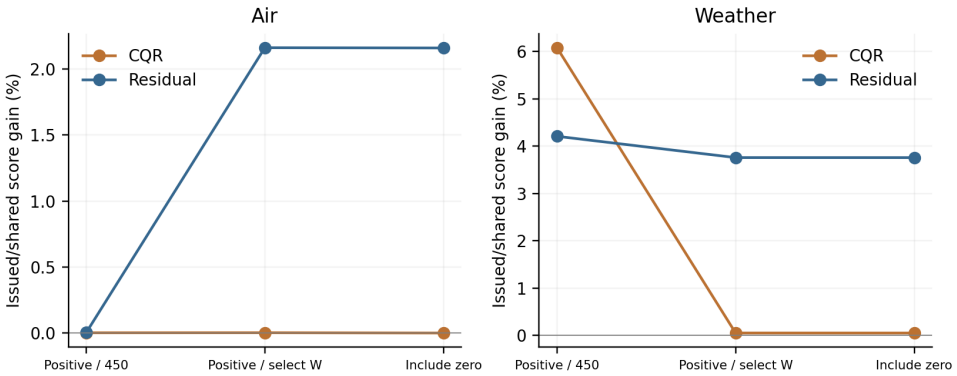


Figure 1: Post-review exploratory contrasts at $h = 24$, four dependent pairs per source. Positive rates include values as small as 10^{-6}. Window selection uses the fixed pre-test validation segment and the existing grid. The weather CQR gap shrinks before zero is admitted; residual-score selection follows a different pattern. These points compare selected configurations, not intrinsic state effects.

2. Related work and comparison scope

CQR uses predictive quantiles to define calibrated interval endpoints [5]; residual scaling supplies another established construction [6]. Online methods include ACI, adaptive aggregation, strongly adaptive experts, PID, decaying-step optimization and optimistic updating [7, 8, 9, 10, 11, 12, 13, 14]. Our primary contrasts hold the neural forecasts fixed while changing these components. Declared implementation variants and their source checks are in Appendix J.

Adler et al. [15], published at ICLR 2026, examine native TSFM calibration, prediction heads and rollout. Distribution-aware conformal prediction [16] already separates probabilistic predictors from scores. We investigate a different comparison: whether changing the score changes the apparent delayed-state benefit after rate and memory controls, and which native-band benefits survive common support and asymmetric residual constructions. Foundation-model conformal inference [17], direct volatility modeling [18] and hidden-state-conditioned calibration [19] provide related approaches; hidden representations are unavailable in the forecast archives used here.

AcMCP models multi-step error autocorrelation and provides ACI/PID implementations [20]. Its ACI uses a shared state. The delayed ACI analysis of El Halabi and Brandt [21] establishes interleaving for issuing-state updates and examines delay relative to residual memory, including rate dependence on residual dynamics. Our window intervention changes the retained empirical score history on the same forecast stream; it does not estimate that latent residual-memory scale. Staggered integral control also treats delayed feedback [22], and O^{2}CP optimizes across horizons [23]. We implement full configured AcMCP-CSS on the complete weather subset and retain the wider PID comparison.

Three comparison types are kept distinct. Information contrasts hold the controller fixed and change the score or interval construction. State contrasts at a fixed window hold information and observation timing fixed, while allowing each state to select its rate. Joint window selection compares deployable configurations; it does not isolate state alone. AcMCP/PID versus native CQR is a complete-method comparison with different information. These distinctions define the scope of the empirical contribution; we do not infer new algorithmic validity guarantees from modified implementations.

3. Causal calibration and explanatory accounting

3.1. Scores, observation timing and state

At origin t, the frozen model emits a point forecast m_t and ordered endpoints ℓ_t, u_t for $y_t = Y_{t+h}$. At natural feedback lag h, calibration can use only outcomes from origins $i \le t-h$. The primary scores and inversions are

$$s_t^{\mathrm{R}} = |y_t - m_t|, \qquad C_t^{\mathrm{R}} = [m_t - q_t, m_t + q_t], \tag{1}$$

$$s_t^{\mathrm{Q}} = \max(\ell_t - y_t, y_t - u_t), \qquad C_t^{\mathrm{Q}} = [\ell_t - q_t, u_t + q_t]. \tag{2}$$

CQR can have a negative threshold and shrink its native band. Reversed endpoints are repaired to their midpoint and counted; coverage feedback uses the emitted interval. Infinite scores remain infinite. The stored 0.1/0.9 band is a score input, not an uncalibrated 90% interval. No unrecorded tails are extrapolated. Additional scale and midpoint controls are specified in Appendix L.

Let $Q_p(\mathcal{H}_t)$ be the linearly interpolated empirical quantile of the last W observed, revealed scores. Rolling uses $q_t = Q_{1-\alpha}(\mathcal{H}_t)$, with $\alpha = 0.1$. ACI uses $q_t = Q_{1-a_t}(\mathcal{H}_t)$. For emitted-interval miscoverage $e_i = \mathbf{1}\{y_i \notin C_i\}$, the

two established recursions are

$$a_t^{\mathrm{S}} = \mathrm{clip}_{[.001,.999]}\{a_{t-1}^{\mathrm{S}} + \gamma(\alpha - e_{t-h})\}, \tag{3}$$

$$a_t^{\mathrm{I}} = \mathrm{clip}_{[.001,.999]}\{a_{t-h}^{\mathrm{I}} + \gamma(\alpha - e_{t-h})\}. \tag{4}$$

They coincide at $h = 1$. At larger h, issued updating has h interleaved level states; the common rolling history still couples their score information. Both start at $a = \alpha$. At $\gamma = 0$ they are the same Rolling method when window and timing agree. Rolling continues to update its empirical distribution even though it does not adjust the nominal level.

All trajectories exclude their first 600 forecast pairs from evaluation. Issuance starts at $600 + h - 1$; validation uses $[600 + h - 1, 1200 - h + 1)$ and testing starts at 1200, with state carried across the boundary. Thus every validation label is available before test selection. Missing neural inputs are filled using past observations only. Missing targets are excluded from score memory, error updates and evaluation; issued phases continue without an increment. The full time-index and missingness checks accompany the code.

3.2. What the accounting explains

Clipping affects the usual telescoping coverage argument. For a revealed-error update, write $r_k = a_{k+1} - a_k - \gamma(\alpha - e_k)$. Then

$$\frac{1}{K}\sum_{k=1}^{K} e_k - \alpha = \frac{a_1 - a_{K+1}}{\gamma K} + \frac{\sum_{k=1}^{K} r_k}{\gamma K}, \qquad \gamma > 0. \tag{5}$$

Bounded levels control the endpoint term but do not eliminate accumulated clipping. For the delayed interpolation with raw update $(1-\rho)a_{t-1} + \rho a_{t-h} + \gamma(\alpha - e_{t-h})$, define $V_t = a_t + \rho\sum_{k=1}^{h-1} a_{t-k}$. Direct subtraction gives

$$V_t - V_{t-1} = \gamma(\alpha - e_{t-h}) + r_t. \tag{6}$$

Shared and issued states correspond to $\rho = 0, 1$. The identity separates finite-time endpoint and clipping contributions; unrevealed final outcomes are counted separately. It is not a proof of conditional validity or a decomposition of all score differences.

Under a stationary differentiable miscoverage response with slope $g > 0$, inactive clipping and deterministic response feedback, linearized shared and issued errors satisfy $x_t = x_{t-1} - \gamma g x_{t-h}$ and $x_t = (1-\gamma g)x_{t-h}$. Their familiar stability ranges are

$$0 < \gamma g < 2\sin\left(\frac{\pi}{4h-2}\right) \quad \text{(shared)}, \qquad 0 < \gamma g < 2 \quad \text{(issued)}. \tag{7}$$

The first is the classical scalar delay-difference boundary [24]; issued contraction per calendar step is $|1-\gamma g|^{1/h}$. A wider stable gain range therefore need not mean faster tracking. These approximations motivate rate controls. Empirical quantile jumps, changing memory and clipping require the actual trajectory diagnostics in Appendix G.

4. Experimental design

4.1. Evidence panels and analysis stages

The original main panel has 31 series and four archived backbones (Chronos-2, Chronos-Bolt, TimesFM and Moirai): 25 observed energy series, four constructed series and two synthetic switches. Separate renewable and multi-horizon archives provide supporting diagnostics. An original confirmation uses traffic and exchange rates with two Chronos models, giving 16 pairs per horizon. These sources, time units and archive limitations are specified in Appendix A.

A further locally fixed protocol precedes numeric acquisition of two new sources: Jena roof temperature and pressure in the first 3,600 hours of 2026, and PM2.5 at two Beijing stations in the first 3,600 hours from 1 March 2013 [25, 26]. The same two cached Chronos checkpoints generate 24 horizons at 2,976 consecutive origins, with context 512. Each test has 1,776 origins before excluding missing targets. Weather forecast targets are complete; air-quality masks are retained. Four series and two models yield eight dependent pairs, not eight independent replications.

Original and new-source confirmation choices were recorded before their respective acquisitions. Later rate sensitivities, PV controls and the present positive-rate, time-block and history analyses revisit inspected data and are exploratory. Local timestamp/hash records are not external preregistration. Neither source novelty within this calibration project nor recent observation dates establish exclusion from neural pretraining. Detailed grids and chronology are in Appendix M.

4.2. Selection and comparisons

The original matched ACI comparison selects each score's (γ, W) from three rates and $W \in \{200, 450, 900\}$. The new-source protocol nests the original three rates, a wider grid with delay-scaled values, zero/small rates down to 10^{-6}, and finally window selection. Each state minimizes its own validation interval score, with ties favoring smaller rate and then window. A separately selected zero-rate Rolling control uses the same candidate windows. Test scores never choose the search range. Positive-rate reanalysis

uses only existing candidates, holding W fixed or selecting it before finally admitting zero.

Full AcMCP-CSS uses the pinned author's MA$(h-1)$ and cross-horizon regression, all 24 horizons, history 600, and validation-selected learning rates. CSS is the author's supported fitting option; its estimates need not equal default CSS-ML. We add history 450 for a bounded sensitivity. Because the author's history parameter also governs startup, this additional comparison uses a common post-startup validation block $[900+h-1, 1200-h+1)$, unchanged tests, and common histories $\{450, 600\}$ for CQR, Rolling and AcMCP/PID. Every evaluated scorecast is checked finite. Fixed-600 and selected-history results answer complementary fairness questions without presenting different fitting windows as equivalent algorithms.

On all 32 original PV pairs, the signed-error comparator uses $[m_t + Q_{a_t/2}(Y-m), m_t + Q_{1-a_t/2}(Y-m)]$ with the same total-miscoverage state. Each construction is tuned before and after common support projection $[L, U] \mapsto [\max(0, L), \max(0, U)]$. Projection leaves each candidate's raw feedback trajectory unchanged, isolating emitted-interval geometry. For a nonnegative target it cannot increase interval score or lose a covered observation. In addition to actual zero/positive groups, an issue-time grouping uses whether m_t exceeds the median of the first 600 forecasts. Its threshold never uses test outcomes.

4.3. Metrics and temporal resolution

For $[L, U]$, we use the proper interval score

$$\mathrm{IS}_\alpha = (U-L) + \frac{2}{\alpha}(L-y)_+ + \frac{2}{\alpha}(y-U)_+, \tag{8}$$

alongside coverage and local absolute coverage error over 150-origin windows [27, 28]. Missing targets are excluded from each window's denominator. Paired score gain is $100(1-\mathrm{IS}_A \,/\, \mathrm{IS}_B)$; ratios and gains are calculated within each model–series pair before averaging, with equal source weight for equal-sized confirmation panels. A positive local-error improvement means the comparator's error minus the candidate's, in percentage points.

Three consecutive 592-origin test blocks reveal temporal heterogeneity without reselecting models. Block-local windows stay within each block. The main-panel bootstrap and dependence checks remain in the supplementary records; confirmation pair counts and tiny score differences are descriptive, not independent-replication confidence statements. We report score, overall coverage and local error together.

5. Results

5.1. RQ1: What predictive-information benefits survive stronger controls?

Across 124 main-panel pairs, default CQR–ACI improves score by 13.57% relative to default symmetric residual–ACI. Tuning both methods reduces the paired benefit to 9.98%. The latter is recomputed from every prediction archive and all nine candidate settings for each score. Native-scale and midpoint controls, the full original method table and family heterogeneity remain in Appendix L and Appendix K; the default CQR benefit is 6.03% after excluding PV. On traffic/exchange rates, holding COP fixed gives native-score benefits of 12.27%, 14.66% and 22.16% at $h = 1, 6, 24$, respectively. Magnitudes depend on the controller and panel; the comparisons do not pool scores in incompatible units.

The PV controls test two simpler explanations for the large original solar result: symmetric intervals and extension into impossible negative outputs. Table 1 shows that the positive-output benefit declines from 12.77% to 11.57% after common projection, and to 9.56% against projected signed residuals. Native information therefore retains value after both controls. Successive percentages have different denominators and selected settings, so their differences are not additive causal contributions. The original zero/positive score decomposition and all width and miss-penalty terms remain in Appendix H and Appendix L.

Table 1: PV construction controls, all 32 pairs, validation-tuned. Gains are mean paired CQR improvements over the named baseline (%); coverages are percent. High forecast is an issue-time observable prefix-median split.

Support	Baseline	Group	CQR gain	Cov. CQR	Cov. base
Raw	Residual	all	20.51	89.72	89.34
Raw	Residual	positive	12.77	86.32	80.17
Raw	Residual	forecast high	14.70	87.03	82.91
Nonnegative	Residual	all	15.61	91.56	90.55
Nonnegative	Residual	positive	11.57	86.45	80.50
Nonnegative	Residual	forecast high	12.41	87.07	83.04
Nonnegative	Signed	all	10.86	91.56	91.40
Nonnegative	Signed	positive	9.56	86.45	80.09
Nonnegative	Signed	forecast high	9.26	87.07	84.59

5.2. RQ2: What does adaptation add after rate and memory selection?

The earlier traffic/exchange-rate sensitivity reduced the CQR issued/shared gain at $h = 24$ from 8.43% to 0.99% after widening the rate range. On the new-source confirmation, small/zero rates remove the

air-quality gap but leave 6.08% on weather at $W = 450$. Adding window selection lowers that gap to about 0.05% (Table 2). This is why a rate-only explanation is incomplete.

Table 2: Locked new-source confirmation at $h = 24$. Mean paired issued/shared score gain (%), four pairs per source. Only the last column selects memory.

Source	Score	Original	Wide	Zero/small	Window
Air quality	Residual	15.04	3.08	0.00	2.16
Air quality	CQR	10.59	0.23	0.00	0.00
Weather	Residual	7.82	4.32	4.21	3.76
Weather	CQR	9.21	6.11	6.08	0.05

Does this attenuation merely select the identical zero-rate method? The post-review positive-rate controls separate those possibilities (Table 3). Weather CQR has a 6.080% gap with positive rates at $W = 450$, 0.052% when positive-rate methods select their window, and 0.052% after zero is admitted at the displayed precision. The window-dependent attenuation therefore occurs without exact zero. However, six of eight issued and seven of eight shared CQR positive-rate selections use 10^{-6}; positivity alone does not establish substantial feedback. The result describes the tested search and its rolling limit.

Table 3: Exploratory positive-rate decomposition, $h = 24$, mean paired state gain (%). The first three columns exclude zero. "Select W" uses 200, 450 and 900; "Add zero" differs only by admitting zero. Tiny rates remain eligible.

Source	Score	$W = 450$	$W = 900$	Select W	Add zero
Air	CQR	0.002	-0.003	0.002	0.000
Air	Residual	0.004	2.161	2.161	2.159
Weather	CQR	6.080	0.052	0.052	0.052
Weather	Residual	4.208	0.001	3.759	3.758

Residual scores show why selected configurations must also be inspected. On weather, their state gap is essentially zero when both use $W = 900$, yet is 3.759% after independent window selection. The shared residual method selects $W = 450$ for both pressure series while the issued method selects 900. The resulting test gap is partly a difference in validation-selected memory, not a residual intrinsic state effect. On air quality, allowing window selection also brings back a residual gap. Larger validation searches need not improve every test result.

Table 4 exposes every final CQR choice. Six of eight pairs choose the exact same zero-rate/window method. The two pressure pairs choose distinct settings and retain small score differences. Seven of eight configurations in each state select the largest allowed window; early and late validation select identical settings for only three issued and four shared pairs. These boundary and stability checks preclude interpreting the final mean as a precisely identified state effect.

Table 4: Every final CQR pair at $h = 24$, original full-validation selector. S/I are shared/issued; C2 and Bolt are the two Chronos models. IS is in the respective series' units. Same rolling means both rates are zero with the same window. Air A/D denote Aotizhongxin/Dingling.

Pair	γ_S	W_S	IS_S	γ_I	W_I	IS_I	Selection
Air A / C2	0	900	288.250	0	900	288.250	Same rolling
Air A / Bolt	0	450	319.915	0	450	319.915	Same rolling
Air D / C2	0	900	238.382	0	900	238.382	Same rolling
Air D / Bolt	0	900	300.438	0	900	300.438	Same rolling
Pressure / C2	0.0001	900	21.442	0.002	900	21.437	Distinct
Pressure / Bolt	0	900	20.385	0.00208	900	20.348	Distinct
Temperature / C2	0	900	15.152	0	900	15.152	Same rolling
Temperature / Bolt	0	900	14.491	0	900	14.491	Same rolling

Against separately window-selected Rolling CQR, the final issued CQR mean score gain is zero on air quality and 0.04% on weather at $h = 24$; weather coverage is 89.43% versus 89.51%. The earlier traffic/exchange-rate comparison likewise gives only 0.17% mean gain over Rolling. COP instead has a 2.90% higher score there but better overall and local coverage. The correct interpretation requires the reliability comparisons in RQ3, rather than a score-only ranking.

A common-history AcMCP check.. The original full AcMCP-CSS comparison used history 600 against CQR allowed to select 200/450/900. At $h = 24$ its mean paired score ratio to selected Rolling was 1.147, with 87.54% coverage. The extra comparison in Table 5 restricts all methods to common history options and a common post-startup validation segment. It includes every weather pair and the no-scorecast PID ablation. With common history 600, AcMCP's score ratio is 1.172; selecting 450/600 gives 1.164, with 89.18% coverage. Fixed 450 scores better on the test than the joint selector, another reminder that validation search is not a test oracle. Issued CQR remains close to Rolling, whereas shared CQR has a 1.039 score ratio under the joint selector. Thus the nearly zero original state gap does not persist under every history and validation choice. The history check is a bounded

complete-configuration sensitivity; native CQR and signed-error methods still receive different predictive information. All horizons, selected rates, success counts and source checks are retained in Appendix N.

Table 5: Common-history weather comparison at $h = 24$, four pairs. All rows use the same post-startup validation block and unchanged test origins. Each IS ratio uses its matched Rolling method within that history search; coverage is percent and Local is percentage points.

History	Method	IS ratio	Coverage	Local
450	Rolling-CQR	1.000	89.32	6.33
450	CQR-shared	1.136	83.22	8.96
450	CQR-issued	1.000	89.27	6.34
450	AcMCP-CSS	1.138	89.18	6.45
450	PID-no-scorecast	1.061	88.20	6.49
600	Rolling-CQR	1.000	89.44	6.75
600	CQR-shared	1.039	87.43	7.42
600	CQR-issued	1.000	89.43	6.74
600	AcMCP-CSS	1.172	89.27	6.27
600	PID-no-scorecast	1.067	88.37	6.43
450/600	Rolling-CQR	1.000	89.44	6.75
450/600	CQR-shared	1.039	87.43	7.42
450/600	CQR-issued	1.000	89.43	6.74
450/600	AcMCP-CSS	1.164	89.18	6.48
450/600	PID-no-scorecast	1.065	88.60	6.24

5.3. RQ3: Which reliability gains and deficits remain?

Adaptive levels can supply local reliability even when average-score improvements are small. On weather at $h = 1$, adaptive CQR improves mean paired score by 0.36% while reducing average local coverage error from 3.15 to 1.43 percentage points. Table 6 shows every pair and every test block. Both pressure series and Chronos-2 temperature improve local error in all three blocks, despite mixed signs for their score gains. Chronos-Bolt temperature selects the same zero-rate rolling configuration and has identical results. This is neither a universal benefit nor an effect of issued versus shared states, which coincide at one step.

The benefit varies with horizon and source (Table 7). The complete block-by-pair display in Appendix N shows where favorable averages conceal different time periods. These descriptive slices identify the empirical extent of the benefit; they do not establish that a measured drift variable caused it. The average interval score rewards width and miss distance, whereas

Table 6: Weather $h = 1$: adaptive CQR versus selected-window Rolling. Gain is paired score improvement (%). Whole-test local errors and their block improvements are percentage points; positive block values favor ACI. All configurations are selected before testing.

Pair	Score gain	Local roll	Local ACI	Early	Middle	Late
Pressure / C2	+1.77	4.51	1.45	+2.83	+3.88	+3.81
Pressure / Bolt	+0.20	3.75	0.86	+2.76	+2.58	+3.01
Temperature / C2	-0.54	2.14	1.19	+1.08	+1.05	+1.15
Temperature / Bolt	+0.00	2.21	2.21	+0.00	+0.00	+0.00

local coverage error penalizes persistent deviations from the target frequency. Improvements on one metric therefore need not have the same size or sign on the other.

The same distinction is visible for another method. In the common-history sensitivity at $h = 1$ and history 600, AcMCP-CSS has a score ratio of 1.023 but local error of 2.09 percentage points, compared with Rolling's 3.44. PID reaches 1.86 points. These use the same later validation block and fixed history, so their local gains are not compared with a differently tuned Rolling denominator. Local-reliability value is therefore not exclusive to native CQR, and a slightly worse average score need not make a feedback configuration uniformly inferior.

Table 7: Whole-test tradeoffs for the original new-source selector, four pairs per source. ACI denotes issued CQR (identical to shared at $h = 1$); gain is relative to selected-window Rolling CQR. Coverage is percent and Local is percentage points.

Source	h	Gain	Cov. roll	Cov. ACI	Local roll	Local ACI
Air	1	-0.77	90.59	90.55	2.32	1.94
Air	6	+0.00	89.65	89.65	4.85	4.85
Air	24	+0.00	89.84	89.84	6.37	6.37
Weather	1	+0.36	89.37	89.44	3.15	1.43
Weather	6	-0.18	90.27	90.44	2.91	2.91
Weather	24	+0.04	89.51	89.43	6.03	6.01

Exact clipping accounting explains one concrete coverage cost in the older traffic/exchange-rate panel. Original-grid shared CQR has 2.150 percentage points of revealed-test excess miscoverage: 2.117 from clipping and 0.033 from the endpoint term. With scaled rates, clipping falls to 0.336 while the endpoint term rises to 0.681. Smaller rates suppress one cost while increasing a finite-time adaptation term. The full identity and controlled stability/tracking simulations are in Appendix G; this accounting does not assign every score change to instability.

PV exposes a different reliability deficit. Under the strongest construc-

tion control in Table 1, CQR covers 91.56% overall but only 86.45% of positive targets. The issue-time high-forecast group has a 9.26% score gain against projected signed residuals, yet only 87.07% coverage. Projection can newly cover zero outputs when both raw endpoints were negative, improving overall coverage without resolving positive-output reliability. Thus the retained native information benefit and the retained operational gap are simultaneous findings.

6. Discussion and limitations

The evidence supports a sequence of evaluation decisions. Establish native-information value under matched controllers and plausible interval geometry. Then compare feedback updates with selected-window Rolling, separating a common-window state contrast from a joint configuration-selection contrast. Finally inspect local and observable-group reliability. This sequence distinguishes three outcomes observed here: surviving native-band benefits, a window-sensitive state-score difference that becomes practically small, and local benefits that average scores understate.

The zero-rate result has a precise interpretation. Six pairs select the same method, so equality is mathematically expected; the empirical finding is that validation chooses that method and it remains competitive on later observations. The positive-only controls show that exact degeneracy is not the sole explanation for the weather attenuation, but tiny selected rates and window boundaries remain important. The residual fixed-window versus selected-window contrast further shows how a favorable apparent state gap can arise from different memory choices. These observations guide fair comparisons without providing a universal controller selection rule.

Generality remains limited. Weather comes from one site and two variables; air quality uses two correlated stations, with two models and relatively short tests. The panels include dependent energy series and constructed switches. New-source choices were locked locally, whereas the positive-only, block and extra history analyses use inspected data. All selectors and negative results are retained, but these sensitivities are not additional independent confirmations. Broader uninspected deployments would be needed to validate a practical adaptation-selection rule.

Finite search boundaries also matter. Maximum-window selections are frequent, and the extra AcMCP comparison covers histories 450/600 with CSS and a fixed integrator configuration. It does not optimize every AcMCP setting or implement O^2CP. Complete-method rankings cannot isolate controller quality. Original historical checkpoint metadata remain incomplete;

exact spot checks and paired forecasts support the calibration comparison but do not prove training-data exclusion. The new archives retain checkpoint hashes and generation details. The default CPU artifact reruns calibration; optional R fitting and neural generation have separate dependencies and declared scope.

Finally, overall or local frequency diagnostics do not establish conditional coverage, and lower interval score does not by itself quantify operational savings. The latter would require a decision policy and explicit shortfall costs. Our contribution is the tested attribution and reliability evidence at the stated scales, with those boundaries visible rather than absorbed into a single method ranking.

7. Conclusion

Native predictive information retains benefits after matched tuning, support and asymmetric-residual controls. Delayed-state score advantages depend on rate and memory selection; the weather gap shrinks under positive-rate window controls before exact zero is admitted. Near-rolling long-horizon scores coexist with one-step local-reliability benefits from adaptive levels, while PV subgroup undercoverage persists. Reporting these distinctions changes how calibration methods should be compared and how their deployment evidence should be interpreted.

Data and code availability

Supplementary Software S1 (`Supplementary_Software_S1.zip`) contains 164 forecast archives, source/protocol hashes, CPU reproduction, reference results and an explicit `ARTIFACT_MAP.md`. The entry point recomputes the 124-pair headline, rate/window searches, PV controls and the new focused analyses. Cached AcMCP replay uses supplied R fits; separate commands refit the full author algorithm. Earlier unrerun controllers remain identified historical inputs. A repository URL will be added after author upload; the prepared archive is the reviewer-access package.

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Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

OpenAI Codex assisted with literature retrieval, code auditing, experiment implementation, analysis and manuscript preparation. The authors remain responsible for reviewing the methods, results, citations and final text.

Appendix A. Dataset and archive details

UCI consumer indices are 5, 37, 70, 103, 136, 168, 201, 234, 267 and 300 in the public LSTNet-format electricity matrix. Alabama plant indices are 3, 21, 39, 57, 75, 93, 111 and 130. Main observed traces use the initial 8,000 raw observations, except the two minute-level ETT channels (12,000) and ERCOT channels (16,000). The ETT columns are ETTh1/OT, ETTh2/OT, ETTm1/HUFL and ETTm2/OT.

The two cross-consumer splices concatenate 1,500-point segments starting at raw offset 1,000 from clients (10,40,70,100) and (25,90,140,210). Seasonal constructions use clients 20 and 80, concatenating 1,200-point windows starting at offsets 4392, 8784, 13128 and 17544. They deliberately omit the intervening chronology. Synthetic series have length 6,000, seeds 1 and 7, AR coefficient 0.7, a sinusoid of period 200 and four piecewise mean/innovation-scale regimes $(0, 1), (0, 2.2), (5, 0.8), (-3, 1.6)$.

The multi-horizon panel uses electricity clients 20 and 80, solar plants 10 and 30, ETTh1/OT, the first cross-consumer splice, seasonal client 20 and synthetic seed 1. Origins stop 24 steps before the raw-series end for every horizon, ensuring the same origins within that panel. Public data sources are the LSTNet data repository[1], ETDataset[2], ERCOT historical load[3] and GEFCom2014.

[1] `https://github.com/laiguokun/multivariate-time-series-data`
[2] `https://github.com/zhouhaoyi/ETDataset`
[3] `https://www.ercot.com/gridinfo/load/load_hist`

Appendix B. Detailed delayed-feedback results

Table B.8: Multi-horizon comparison. Gains use the residual–ACI score at the same horizon. Local error is in percentage points. Tuning never uses test labels.

Horizon / method	Gain (%)	Cov. (%)	Local (pp)
1 / residual/aci	0.00	89.99	1.77
1 / native-scale/aci	12.91	90.01	1.31
1 / cqr/aci	14.35	90.03	1.39
1 / cqr/pi-log	14.94	90.02	1.50
1 / cqr/saocp	12.48	90.69	2.10
1 / cqr/faci	15.42	89.61	1.68
1 / cqr/aci/tuned	14.87	90.09	1.97
6 / residual/aci	0.00	89.46	2.44
6 / native-scale/aci	-11.77	89.64	1.96
6 / cqr/aci	13.89	89.54	2.06
6 / cqr/pi-log	16.21	90.00	2.53
6 / cqr/saocp	5.26	84.09	6.13
6 / cqr/faci	15.50	86.74	3.81
6 / cqr/aci/tuned	15.47	89.74	3.13
24 / residual/aci	0.00	87.10	4.90
24 / native-scale/aci	-52.27	87.32	4.50
24 / cqr/aci	13.15	87.34	4.52
24 / cqr/pi-log	17.09	89.96	4.86
24 / cqr/saocp	6.56	79.98	10.16
24 / cqr/faci	15.57	82.72	7.94
24 / cqr/aci/tuned	11.62	88.09	5.46

Appendix C. Per-series primary comparison

Table C.9: Default CQR–ACI compared with default residual–ACI, averaged over four backbones.

Series	Gain (%)	Coverage (%)
ETTh1_OT	3.12	89.98
ETTh2_OT	8.27	89.99
ETTm1_HUFL	8.74	89.99
ETTm2_OT	3.70	89.97
electricity_c103	5.21	89.97
electricity_c136	4.87	89.99
electricity_c168	5.90	89.93
electricity_c201	5.33	89.96
electricity_c234	6.61	90.00
electricity_c267	4.74	90.01
electricity_c300	17.21	89.91
electricity_c37	8.05	89.97
electricity_c5	8.43	89.95
electricity_c70	5.72	89.89
ercot_COAST	0.77	90.01
ercot_ERCOT	-1.53	90.00
ercot_NORTH	1.86	89.97
seasonal_elec_c20	6.25	90.09
seasonal_elec_c80	10.90	90.00
solar_p111	34.92	89.90
solar_p130	34.37	89.92
solar_p21	36.12	89.97
solar_p3	35.42	89.97
solar_p39	35.01	90.07
solar_p57	35.63	89.94
solar_p75	36.33	89.98
solar_p93	34.21	89.97
spliced_electricity_4seg	11.36	90.10
spliced_electricity_4seg_b	13.27	90.02
synthetic_T6000	0.31	89.99
synthetic_T6000_s7	-0.42	90.02

Appendix D. Confirmation controller diagnostics

Table D.10: Confirmation contrasts at $h = 24$. Each row contains four series and two models. Gain and median are paired interval-score improvements (%); A is the first method in the comparison and B the second. Coverage is in percent, target 90%. Correlated pairs are retained without interpreting the win count as independent replications.

Comparison	Family	Gain	Median	Wins	Cov. A	Cov. B
Native / residual, COP	FX	11.10	2.73	7/8	89.76	89.72
Native / residual, COP	Traffic	33.22	31.79	8/8	90.45	91.40
Native / residual, issued ACI	FX	3.79	1.06	8/8	89.03	89.10
Native / residual, issued ACI	Traffic	13.20	11.92	8/8	89.83	90.20
Issued / shared, CQR	FX	8.62	5.01	8/8	89.03	86.12
Issued / shared, CQR	Traffic	8.24	7.98	8/8	89.83	89.63

Table D.11: Unified comparison on the revisited panel at $h = 24$. IS ratio averages per-pair interval scores divided by that pair's residual rolling-quantile score; lower is better. Family coverage is percent (target 90%). Local is the family-balanced 150-origin absolute coverage error in percentage points; Singleton is percent. Original rows retain locked-grid results, wide ACI and multi-step PID are subsequent sensitivities. PI-log and SF-OGD retain fixed defaults, not tuned optima.

Score	Controller	IS ratio	FX cov.	Traffic cov.	Local	Singleton
Residual	Rolling	1.000	87.48	89.58	6.77	0.000
Residual	Shared, original	1.063	87.75	88.51	5.66	0.000
Residual	Issued, original	0.929	89.10	90.20	6.58	0.000
Residual	Shared, wide	1.066	87.57	88.07	6.14	0.000
Residual	Issued, wide	0.939	88.53	89.72	6.57	0.000
Residual	PI-log	2.341	89.92	90.03	6.36	2.873
Residual	SF-OGD	5.146	90.18	90.03	6.63	4.356
Residual	COP	1.177	89.72	91.40	6.39	0.249
Residual	Grouped ACI	0.984	86.76	89.23	5.74	0.000
Residual	Fixed-scale OGD	1.096	86.11	90.50	7.14	0.344
Residual	Implicit-delay	1.057	86.05	90.03	6.98	0.000
CQR	Rolling	0.852	87.78	89.41	6.37	0.038
CQR	Shared, original	0.937	86.12	89.63	5.79	0.167
CQR	Issued, original	0.853	89.03	89.83	6.12	0.025
CQR	Shared, wide	0.860	88.02	89.53	6.21	0.031
CQR	Issued, wide	0.851	88.14	89.88	6.22	0.044
CQR	PI-log	2.048	90.04	89.98	6.06	3.011
CQR	SF-OGD	4.528	90.17	90.09	6.47	4.377
CQR	COP	0.879	89.76	90.45	5.57	0.028
CQR	Grouped ACI	0.904	87.45	89.67	5.62	0.018
CQR	Fixed-scale OGD	0.884	88.25	90.92	6.13	0.126
CQR	Implicit-delay	0.884	88.33	90.92	6.07	0.069
Signed error	Multi-step PI	1.163	86.89	89.22	6.68	0.024
Signed error	Multi-step PID	1.176	86.85	89.21	6.92	0.065

Appendix E. Algorithm and integrity checks

The primary evidence concerns the archived forecasts, not a reranking of freshly downloaded checkpoints. Offline spot checks reproduce the point and five quantile outputs exactly at 21 selected origins per model across three main-panel series, for 84 checked model–origin cases. For the two Chronos models, matching the original default batch grouping of 256 contexts resolves differences observed with a seven-context batch. TimesFM's checked outputs match the local 2.0 checkpoint through its Hugging Face adapter; Moirai matches after its original quantile-rearrangement step. Current checkpoint hashes and environment versions are recorded. These checks support archive provenance but do not constitute a rerun of every forecast, and original checkpoint commits were not stored with every archive. Possible pretraining exposure to public benchmark data also precludes an assertion that all test series were unseen during pretraining.

The executable checks test future-target perturbations at lags 1, 6 and 24, identical evaluation masks, unit rescaling, exact stored-level selection, empirical quantile agreement, interval-score arithmetic, finite behavior under zero-width forecasts and the reference SAOCP expert recursion. Data validation independently reconstructs target values from local raw-series loaders. Cache records include archive, protocol and code hashes; aggregations reject stale code/protocol outputs.

Source implementations and licenses are retained for SAOCP/FACI and PID. The SAOCP numerical check covers its nonnegative-score expert recursion, without validating a theorem for signed or delayed extensions. Appendix J specifies controller conventions; the artifact retains complete grids, failures and tail misses. Recorded runtimes include possible compilation overhead and do not measure neural inference speed.

Additional checks cover clipping identities, COP source-recursion equivalence, target/issue-time conversion, grouped feedback assignment and implicit proximal optimality against an independent optimizer and subgradient conditions. Protocols and exploratory amendments remain separately identifiable.

The asymmetric multi-step PID comparison matches the unmodified author implementation over 24 horizons and four configured scorecasters, including future-target perturbations. Check records and tolerances accompany the artifact. The later full AcMCP-CSS weather comparison has 96 complete-trajectory checks and 60 independent scorecast refits.

Appendix F. Matched implicit-update ablation

The implicit ablation exactly solves the proximal objective already considered by Hu et al. [14]. With $p = 1-\alpha$, pinball loss $\ell_p(u) = u(p-\mathbf{1}\{u < 0\})$, and a fixed prefix-scaled step $\eta = \gamma(\max_{i<600} s_i - \min_{i<600} s_i)$ (range floored at 10^{-12}), it uses

$$b_t = b_{t-1} + \eta(e_{t-h} - \alpha), \qquad q_t = \arg\min_q \left\{ \frac{(q-b_t)^2}{2} + h\eta\, \widehat{\mathbb{E}}_t[\ell_p(S-q)] \right\}. \tag{F.1}$$

The same 300-score convention supplies the empirical distribution. The matched OGD ablation sets $q_t = b_t$. Both select from $\{0.005, 0.02, 0.05, 0.1\}$ and these values divided by h. The point of this control is to distinguish an implicit correction's incremental value from the fixed scale and learning rate. The development screen also retains half-step and explicit corrections; it did not validate a new algorithm.

The delay-scaled implicit correction in Equation (F.1) has a plausible stationary rationale. If the true and proxy CDFs have local densities $g > 0$ and $\widehat{g} \geq 0$, its deterministic linearized delayed gain is $c = \eta g/(1+h\eta\widehat{g})$. With a correct slope it is below $1/h$ and hence inside the classical shared-state stability boundary in Equation (7). For fixed slopes, however, this is simply a smaller effective gain. It predicts no automatic advantage over suitably tuned basic gradient updating, and the argument does not cover changing empirical distributions or atoms.

The development screen evaluates 3,200 configuration instances, retaining every setting. At $h = 24$, implicit-delay CQR improves interval score by 2.54% over issued ACI and 2.08% over delay-tuned COP, but only 0.23% over matched fixed-scale OGD. It fails the locally recorded continuation criterion of at least 3% against both strong baselines, even before further validation. On the subsequent confirmation panel, its matched-OGD gain is -0.08%: -0.30% for exchange rates and 0.14% for traffic, with seven of 16 positive differences. We therefore retain this result as a negative ablation, not a new main method. A more sophisticated update formula is insufficient evidence of additional predictive information or better adaptation.

Appendix G. Rate, delay and clipping diagnostics

Figure G.2 uses the common ten-rate subset of the wide grid to display the rate–delay sensitivity. It shows every listed rate's test behavior, not a test-based selector. The original selected shared-state CQR score at each

horizon is a common per-pair denominator for both structures. The additional delay-scaled rates also participate in selection and remain in the full result files.

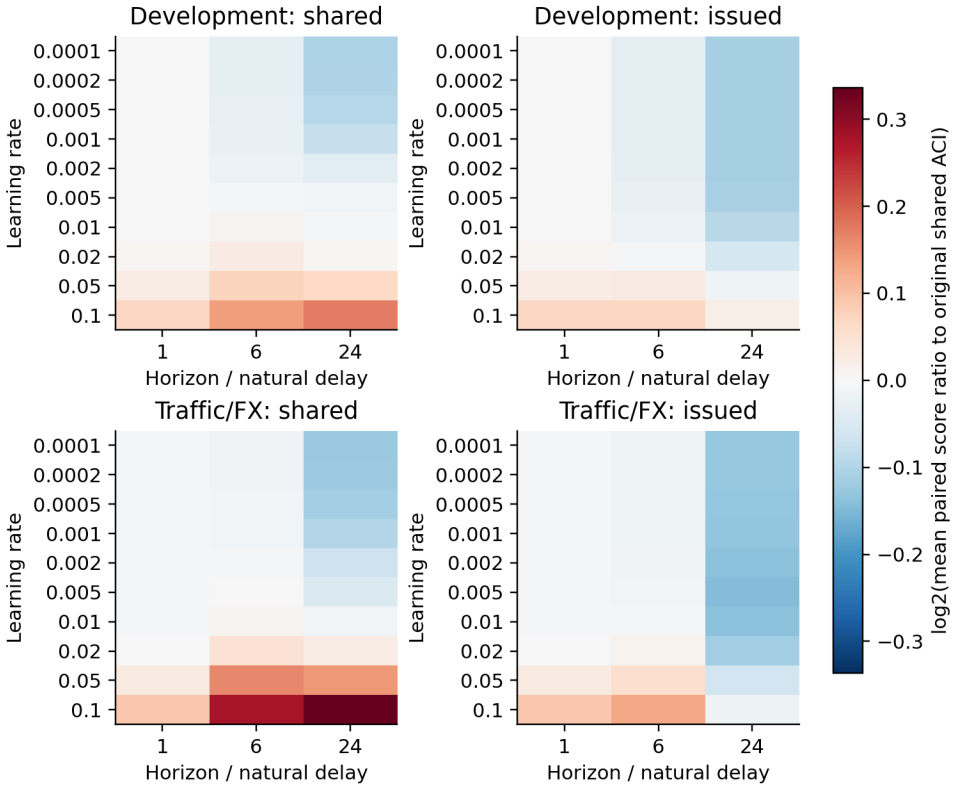


Figure G.2: CQR rate–delay surfaces. Color is the base-two logarithm of the mean paired interval-score ratio to original-grid shared CQR; zero means equal score. Positive values are worse. Development and revisited panels have 16 pairs each. Validation selects configurations; the plotted test surface is descriptive.

For the controlled simulations, the response is prescribed as $M_t(a) = \text{clip}\{0.1 + g(a - a_t^*), 0, 1\}$, with $g \in \{1, 4\}$ and $h \in \{1, 6, 24\}$. Runs contain 12,000 steps. Stationary runs use effective gains at 0.5 and 1.2 times the shared linear boundary. Tracking runs use $a_t^* = 0.1, 0.2, 0.06$ in three 4,000-step segments and the original three rates plus their delay-scaled versions. Deterministic response feedback and Bernoulli feedback are kept distinct, with 20 fixed seeds for the latter, yielding 1,764 controller runs. In the deterministic $h = 24, g = 4, \gamma = 0.005$ tracking example, mean absolute state error over the first 500 steps after both changes is 0.0198 for shared

Table G.12: Exact revealed-test miscoverage decomposition, CQR, $h = 24$, in percentage points. Clip plus Endpoint equals Excess up to rounding. Each interval was issued in the test segment and revealed by the last archive origin. The last 24 issued test intervals are pending; their separately recorded contribution bounds the difference from full issued-test coverage by $24/N_{\text{test}}$.

Panel	Grid	State	Clip	Endpoint	Excess
Development	original	shared	1.801	0.004	1.804
Development	original	issued	0.023	0.393	0.415
Development	scaled	shared	0.043	0.398	0.441
Development	scaled	issued	0.023	0.611	0.633
Development	wide	shared	0.043	0.487	0.530
Development	wide	issued	0.023	0.642	0.664
Traffic/FX	original	shared	2.117	0.033	2.150
Traffic/FX	original	issued	0.135	0.461	0.596
Traffic/FX	scaled	shared	0.336	0.681	1.017
Traffic/FX	scaled	issued	0.135	0.863	0.998
Traffic/FX	wide	shared	0.336	0.909	1.245
Traffic/FX	wide	issued	0.182	0.834	1.016

and 0.0978 for issued updating. This is a specified-response illustration, not a forecast score or a causal diagnosis of all empirical losses.

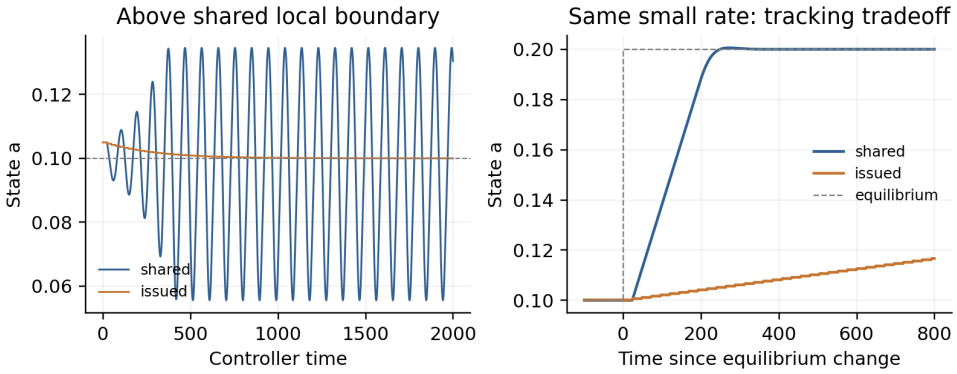


Figure G.3: Deterministic response illustrations at $h = 24, g = 4$. Left: effective gain 1.2 times the shared local boundary. Right: $\gamma = 0.005$, below the boundary for both states, following an equilibrium change. State clipping and response saturation are explicit; the trajectories illustrate stability and tracking, not coverage guarantees.

Appendix H. Photovoltaic score components

Each pair's all-observation score and coverage is reconstructed by weighting its zero- and positive-output components by their own sample fractions. This check is performed before any averaging across plants and models.

Table H.13: One-step PV interval-score components, averaged over the original 32 pairs. IS equals Width plus Lower plus Upper, where the last two are the $2/\alpha$-weighted miss-distance penalties. These means use the shared PV data units; the headline relative gains are instead calculated per pair before averaging. All outcome-defined groups are evaluation-only.

Settings	Score	Group	IS	Width	Lower	Upper
default	cqr	all	2.629	1.766	0.555	0.308
default	cqr	positive	5.978	3.831	1.386	0.761
default	cqr	zero	0.498	0.454	0.025	0.020
default	residual	all	4.046	2.284	1.218	0.544
default	residual	positive	7.607	3.086	3.126	1.395
default	residual	zero	1.781	1.773	0.006	0.002
tuned	cqr	all	2.575	1.816	0.476	0.282
tuned	cqr	positive	5.888	4.017	1.183	0.688
tuned	cqr	zero	0.467	0.417	0.027	0.023
tuned	residual	all	3.240	1.987	0.820	0.432
tuned	residual	positive	6.775	3.626	2.066	1.082
tuned	residual	zero	0.991	0.945	0.027	0.019

There are no negative targets in this evaluated PV panel. The original default and validation-tuned overall scores are reproduced from the archived predictions before decomposition.

Appendix I. Additional sensitivities

Appendix I.1. Scale mismatch can amplify interval widths

Normalization is helpful when the predicted scale tracks error size. It can also amplify historical mistakes. For positive scales, a native-normalized residual interval has width

$$2\tilde{b}_t \, Q_{1-a_t}\Big(\{|y_i - m_i|/\tilde{b}_i : i \in \mathcal{H}_t\}\Big). \tag{I.1}$$

If more than the upper-tail fraction of the calibration history has error magnitude c and scale ε, its empirical threshold can approach c/ε. A later scale B then produces width $2Bc/\varepsilon$. A relative floor keeps the expression finite, but does not eliminate a large historical-to-current scale ratio. Additive CQR does not have this particular multiplicative amplification mechanism, although it can still perform poorly when the base endpoints or update rule are unsuitable.

This argument does not establish that every normalization loss is caused by exact zero bands. Both scale transitions and very small but nonzero

widths matter. We therefore retain all failures, record floor activation and score extremes, and avoid a test-dependent score cap. Nor does the argument establish that a proposed regularizer must improve prediction: the validation and test experiments must decide that question.

Appendix I.2. Normalization failures and unsuccessful selection

At $h = 24$, native-scale ACI has a mean gain of -52.27%, despite a 12.91% gain on the same panel at $h = 1$. Large losses occur on the Chronos-Bolt solar series and on constructed scale switches, as well as smaller losses elsewhere. For example, its interval-score increase on Chronos-Bolt/solar-p10 is 286.35% relative to residual–ACI. These cases are consistent with historical-to-current scale mismatch in Equation (I.1); exact zero bands are not required. The artifacts include width-floor activation rates and normalized-score extremes for every trace.

Regularized normalized CQR does not reliably fix the problem. Across all three horizons, its mean gains range from -12.50% at $\lambda = 0$ to -9.65% at $\lambda = 1$ for the reported endpoint choices, compared with 13.80% for additive CQR–ACI. The validation selector increases the main-panel gain from 13.57% to 13.72%, but its multi-horizon mean is 13.36%, below fixed CQR–ACI. We retain this failed hypothesis instead of expanding the selector grid on the test results. Prefix-based selection remains causal, but a causal procedure need not transfer well across later regimes.

Appendix I.3. Nominal levels, rearrangement and native-output provenance

The primary CQR–ACI benefit persists at 80% and 95% calibrated targets (Table I.14). Omitting quantile rearrangement changes the primary mean gain from 13.57% to 13.56%. The main conclusion is therefore not driven by sorting, although sorting matters more for some individual normalized-score configurations.

Table I.14: CQR–ACI sensitivity, using default residual–ACI at the same nominal level as the denominator. Dense archives are a repeated-output sensitivity, not new independent data.

Panel / miscoverage	Pairs	Gain (%)	Cov. (%)
main-a0.2	124	9.85	80.01
main-a0.1	124	13.57	89.98
main-a0.05	124	14.00	94.90
dense-a0.1	93	13.80	89.98
gefcom-a0.1	39	19.63	89.99

The genuine stored 0.1–0.9 native bands achieve average main-panel coverage of 81.89% (Chronos-2), 87.04% (Chronos-Bolt), 81.96% (Moirai) and 77.57% (TimesFM), against their 80% nominal mass. These values include both overcoverage and undercoverage. They cannot substantiate a claim that all models' native 90% intervals under-cover.

The dense archive requires additional care. TimesFM still stores only the five original levels, so it supplies no genuine 0.05–0.95 comparison. Chronos-Bolt's stored 0.05 and 0.95 outputs equal its 0.1 and 0.9 outputs on approximately 99.86% of rows after rearrangement; requesting more levels did not create independent tail information. Chronos-2's stored 0.05–0.95 band attains 91.69% coverage at a nominal 90%. We therefore do not use the dense files to claim exact full-distribution CRPS or a universal native 90% calibration result.

Appendix J. Controller implementation details

Appendix J.1. One-step and delayed rule variants

We compare seven rules using the same score streams and observation schedule.

1. *Split*: a fixed empirical $(1-\alpha)$ quantile of the initial 600 scores.
2. *Rolling*: an empirical $(1-\alpha)$ quantile of the latest $W = 450$ revealed scores.
3. *Clipped ACI*: replace α in the rolling quantile by a_t, updated as

$$a_t = \mathrm{clip}_{[0.001,0.999]}\{a_{t-1} + \gamma(\alpha - e_{t-d})\}, \qquad \gamma = 0.02, \tag{J.1}$$

   where $e_i = \mathbf{1}\{y_i \notin C_i\}$ and the update is applied only after an issued interval's outcome becomes available. The equation suppresses idle steps before that event.
4. *SF-OGD adaptation*: with $M = \max_{i<600} |s_i|$, update the threshold by $q \leftarrow q - Mg/\sqrt{\sum g^2}$, where $g = \alpha - e$. We initialize at the prefix quantile and allow signed thresholds. This coverage-gradient implementation is described explicitly rather than claimed identical to every tie/projection convention in the published wrapper [10].
5. *PI-log adaptation*: a proportional threshold update and logarithmic tangent integrator inspired by Angelopoulos et al. [11]. With k revealed post-initialization errors and $E_k = \sum_{j=1}^{k}(e_j - \alpha)$,

$$z_k = z_{k-1} + \gamma M(e_k - \alpha), \qquad q_k = z_k + M \tan\left(E_k \frac{\log(k+2)}{k+1}\right). \tag{J.2}$$

The tangent saturates to signed infinity outside $(-\pi/2, \pi/2)$. The initial z is the prefix quantile. We use a fixed prefix scale, omit score-casting and retain our causal delay convention; this is not full reference PID.

6. *SAOCP adaptation*: the strongly adaptive coin-betting expert update from the author implementation [10], with default lifetime multiplier 8 and prefix scale $\sqrt{3}M$. The accelerated residual-score implementation agrees numerically with an independent driver of the original expert class. For signed CQR we remove its nonnegative projection; with delayed feedback, experts process scores in reveal order. These are declared extensions, without transferring the original theorem to the modified setting.
7. *FACI-R*: the author's FACI aggregation update with eight learning rates $0.001\,2^j$, $j = 0, \ldots, 7$, fixed-share mixing and a rolling 450-score empirical survival function. Unlike a full-history wrapper, this variant deliberately shares the rolling history design. Its weighted level is inverted using the empirical inverted-CDF quantile.

The first five rules use linear empirical quantile interpolation where applicable. Consequently, "split" denotes the fixed-threshold comparator here; we do not assert the exact finite-sample rank correction or exchangeability guarantee of textbook split conformal prediction for these dependent streams.

For ACI, we separately select (γ, W) from $\{0.005, 0.02, 0.05\} \times \{200, 450, 900\}$ using only the validation prefix. The score selector retains the declared default rule settings. We report fixed defaults and tuned controls separately, rather than describing one untuned baseline as optimal.

Appendix J.2. Optimistic and groupwise controls

The exact COP reference recursion is checked numerically against the author's implementation. Our delayed adaptation is distinct: at issue t, with $j = t - h$, it uses the preceding 300 revealed scores, excluding s_j, to form the empirical CDF $\widehat{F}_t$ and step scale $R_t = \max s - \min s$. After feedback starts, it updates

$$b_t = b_{t-1} + \eta_t(e_j - \alpha), \qquad q_t = b_t - \tfrac{1}{2}\eta_t\{\widehat{F}_t(b_t) - (1-\alpha)\}, \quad \eta_t = \ell R_t. \tag{J.3}$$

It starts from the 0.9 quantile of the 600-score prefix; before the first issued interval's feedback arrives, the threshold stays at that initial value. Validation selects ℓ from $\{0.05, 0.1, 0.5, 1\}$ and those values divided by h. The divided-rate extension was added after inspecting the development panel and

is explicitly exploratory there. It was fixed before the confirmation evaluation. Singleton repairs use actual emitted-interval feedback; the original COP guarantees are not asserted for this delayed, repaired adaptation.

Grouped ACI is an established groupwise-calibration control [29]. The grouping feature is native band width, with a threshold fixed at its median in the first 600 forecasts; ties enter the low-width group. The development factorial separates score memory, feedback state, both, or neither. Group memory retains 450 scores per group and falls back to global history with fewer than 20 observations. Each revealed error updates the state assigned at its issue time. Confirmation uses the version with both memory and state separated, with the same ACI rate grid.

The matched implicit-update control and its negative result are specified in Appendix F.

Appendix J.3. Configured multi-step PID

We also compare the asymmetric multi-step PID implementation supplied with AcMCP [20, 11]. Its two tails target $\alpha/2$, retain the author's zero internal initialization and warmup recursion, and use error-range-scaled steps and the unbounded tangent integral correction. We set rolling history to 600, $C_{\text{sat}} = 0.5$, and initialize K_I to the maximum absolute error revealed in the first 600 target-clock rows across horizons 1–24, rather than using the whole trace. Validation selects the rate from $\{0.005, 0.01, 0.05, 0.1\}$ and those rates divided by h, jointly with no scorecaster, a rolling mean, the author's documented naive scorecaster, or a declared least-squares AR(1). The latter clips the fitted coefficient to $[-0.995, 0.995]$, recalculates the intercept from sample means, and iterates h steps. It is a configured scorecaster, not the author's default or an AcMCP substitute.

An accelerated implementation reproduces the unmodified author recursion on one 1,000-origin electricity archive for all 24 horizons and four scorecasters, and passes future-target perturbation checks. The benchmark evaluates all 32 development/revisited pairs and all three horizons, totaling 2,560 configurations. Crossed endpoints are collapsed to their midpoint for evaluation, while the internal tail-score feedback is preserved for source fidelity. Infinite or invalid intervals receive infinite score rather than being omitted. A separately selected no-scorecaster control is retained.

Appendix K. Earlier family, rule and delay diagnostics

Appendix K.1. One-step rule diagnostics

Across the seven default rules, adding native information generally improves efficiency, but the effects are not orders of magnitude apart (Figure K.4). FACI-R with residuals already gains 8.57% and has lower coverage, while residual SAOCP overcovers on average. CQR–PI-log, CQR–SF-OGD and CQR–ACI have similar average interval scores but different local errors. SF-OGD has the smallest mean local error among these three one-step CQR variants. A universal assertion that the score renders the rule irrelevant is therefore unsupported even before adding delayed feedback.

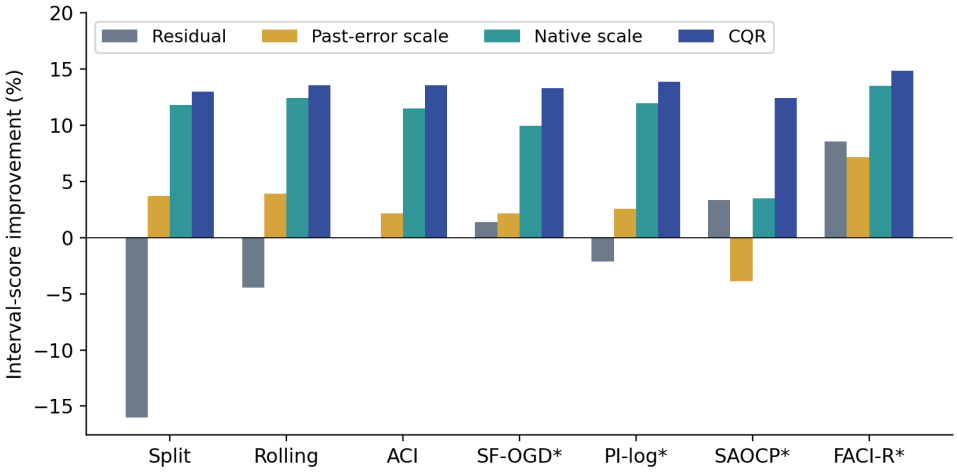


Figure K.4: Score–rule comparison on fixed one-step forecasts. Every gain uses the same default residual–ACI denominator. Asterisks indicate the explicit adaptations described in Appendix J; these labels do not claim equivalence to full original forecasting pipelines.

Appendix K.2. Benefits vary by family and persist on separate renewable series

The mean CQR–ACI gain is 35.25% on Alabama PV, 7.21% on UCI loads, 5.96% on ETT and only 0.37% on the three ERCOT loads (Table K.15). The synthetic mean is slightly negative. This heterogeneity is more informative than a pooled correlation between a post-hoc volatility index and gains. Removing all eight PV series leaves a 6.03% gain, with descriptive interval $[4.25, 7.88]$%. Restricting to the 25 observed series yields 15.16%; that subset gives PV a larger relative weight. Leave-one-family-out means remain positive, ranging from 6.03% to 16.60%.

Table K.15: CQR–ACI gains by main-panel family. Intervals resample series within the listed family and are descriptive; narrow intervals for correlated PV plants do not imply broad geographic certainty. Small two- or three-series groups have especially limited precision.

Family	Series	Gain (%)	Descriptive 95% CI
Alabama PV	8	35.25	[34.75, 35.74]
Constructed seasons	2	8.58	[6.25, 10.90]
Constructed switches	2	12.31	[11.36, 13.27]
ERCOT	3	0.37	[-1.53, 1.86]
ETT	4	5.96	[3.41, 8.51]
Synthetic	2	-0.05	[-0.42, 0.31]
UCI load	10	7.21	[5.53, 9.70]

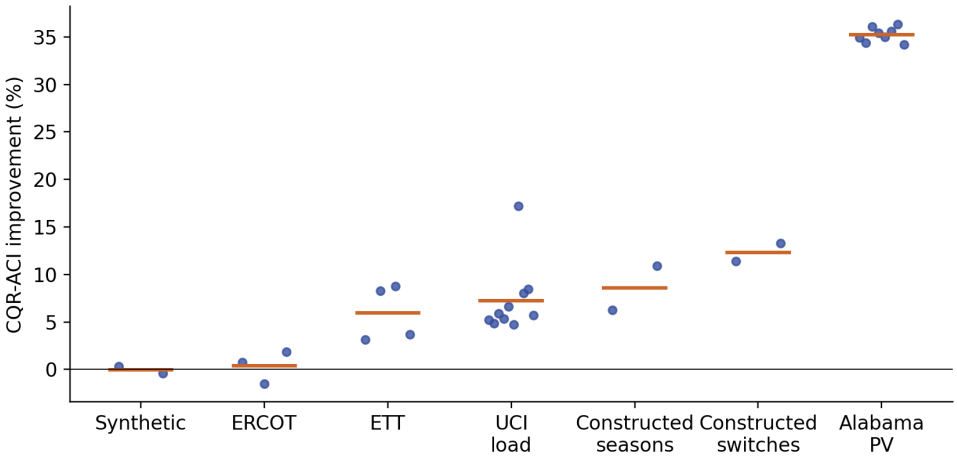


Figure K.5: Each point is one series averaged across four backbones; horizontal orange marks are family means. Constructed and synthetic series remain visible rather than being counted as independent observed energy systems.

Per-backbone mean gains are 15.15% for Chronos-2, 13.82% for Chronos-Bolt, 12.90% for Moirai and 12.41% for the archived TimesFM configuration. On the separate GEFCom panel, CQR–ACI gains 19.63% overall: 41.46% for solar and 13.08% for wind, with 89.99% mean coverage. This is evidence beyond the Alabama solar panel, but does not convert correlated zones into independent geographic samples.

At block lengths 100, 200 and 400, 109 primary pairs have a positive paired-bootstrap 95% interval and two have a negative interval. The centered-test/BH procedure identifies 109 positive and two negative differences at each of these block lengths. These counts are reported with the dependence caveat above; the magnitude and family distribution of the effect remain the main empirical evidence.

Appendix K.3. Delayed feedback makes rule choice consequential

Figure K.6 and Table B.8 show the joint horizon/feedback experiment. At $h = 24$, CQR–ACI has 87.34% mean coverage, compared with 87.10% for residual–ACI. CQR–PI-log attains 89.96% coverage and a 17.09% score improvement relative to residual–ACI, compared with 13.15% for CQR–ACI. CQR–ACI's average number of clipped updates, counting validation and test adaptation, increases from 3.06 at $h = 1$ to 201 at $h = 24$. This is consistent with the projection term in Equation (5), although a count alone does not measure its signed magnitude.

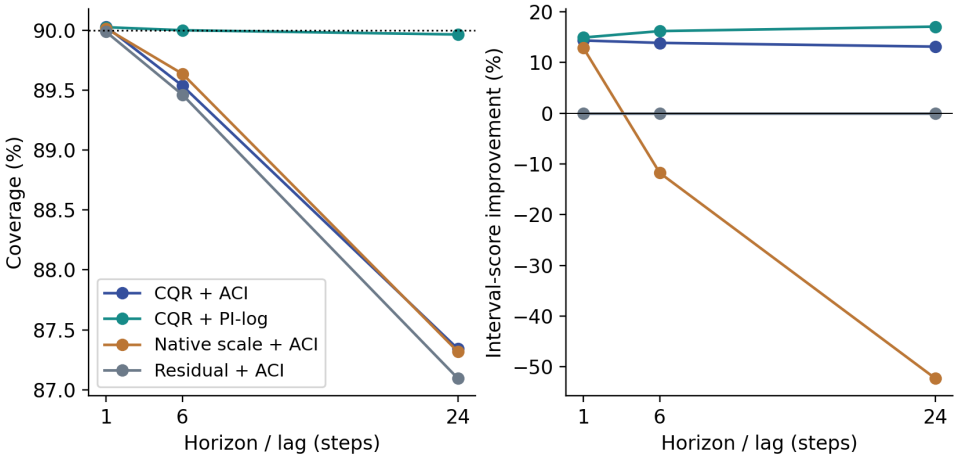


Figure K.6: Joint changes in horizon and natural feedback delay on eight series and two Chronos models. Coverage is an average; score gains use the horizon-specific residual–ACI denominator. Solar steps are ten minutes, so $h = 24$ is not day-ahead for those series.

The PI result is not a claim of uniformly improved reliability. At $h = 24$, its local 150-origin coverage error is 4.86 percentage points, versus 4.52 for CQR–ACI. Improved overall coverage can coexist with worse local variation. The controller also emits some singletons after negative signed corrections: CQR–PI-log's average singleton rate is 0.567% at $h = 24$, versus 0.048% for CQR–ACI. No infinite test intervals occur for these two reported variants in this panel.

The fixed-forecast intervention separates lag from changing forecast difficulty (Figure K.7). On the same $h = 1$ predictions, increasing calibration lag from 1 to 24 steps changes residual–ACI coverage from 89.99% to 88.62% and CQR–ACI from 90.03% to 89.52%. CQR–PI-log remains at 89.99% for lag 24, while its local coverage error is 2.88 percentage points compared with 2.08 for CQR–ACI. Thus feedback restriction alone can alter the relevant coverage comparison. The larger deficit in true $h = 24$ forecasts additionally reflects changes in the forecast/score stream; lag alone does not explain its entire magnitude.

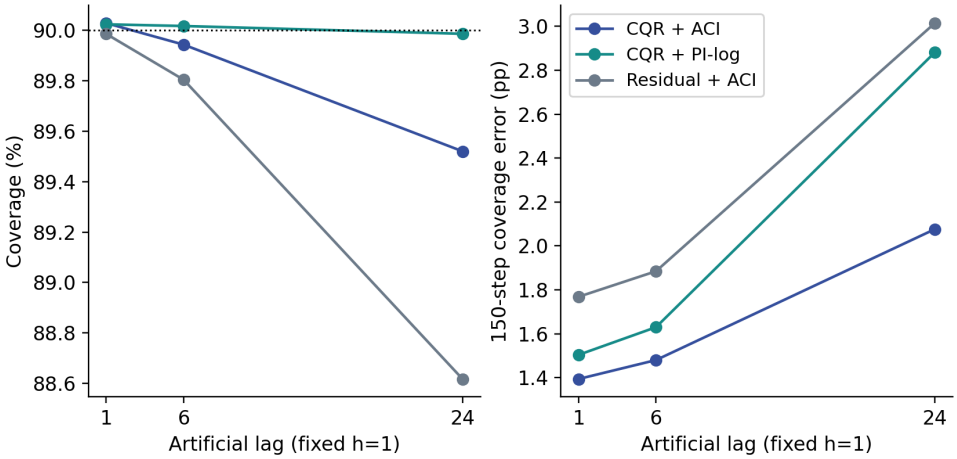


Figure K.7: Exploratory intervention holding one-step predictions and targets fixed while withholding calibration outcomes. Overall coverage and local coverage error respond differently. This isolates the calibrator's feedback channel, not the model's context.

Appendix K.4. Earlier long-horizon PV diagnostics

The separate long-horizon diagnostic concerns two other development photovoltaic series and two backbones, so it is not a within-panel comparison with the one-step decomposition. On the two development photovoltaic

series and two backbones, tuned CQR–COP at $h = 24$ attains 90.41% overall coverage, yet only 79.93% on positive-output targets and 97.05% on zero-output targets (Figure K.8). Zero outputs constitute approximately 61.04% of these test segments. At $h = 1$ and $h = 6$, positive-output coverage is 87.95% and 82.42%, respectively, despite overall coverage near 90%. These are four model–series pairs from two series, not four independent solar datasets. Positive output is an outcome-defined diagnostic and need not coincide with astronomical daytime. A marginal coverage theorem does not promise coverage on this subset, so the finding is not a counterexample to that theorem.

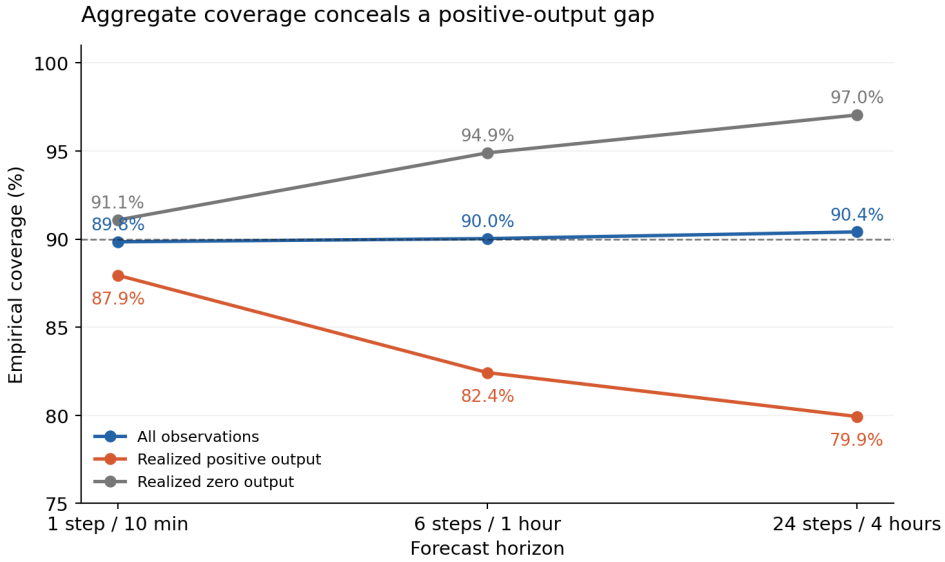


Figure K.8: Post-hoc photovoltaic reliability diagnostic for selected CQR–COP. Coverages average over two series and two models. The zero/positive labels are used only for evaluation; prediction and validation never use the future target label. At this sampling rate, 24 steps represent four hours.

The observable-width grouping experiment tests whether a simple existing approach can reduce the deficit. Separating both memory and state increases positive-output coverage from 78.93% to 83.49% relative to matched shared-state CQR on these solar pairs. Overall coverage changes from 88.46% to 89.45%, with a 2.47% interval-score gain. The positive-output gap remains substantial. Separating memory alone improves solar interval score by 6.75% but reduces overall coverage to 85.66%, illustrating why a

proper-score change should be accompanied by coverage diagnostics. Across all 16 development pairs, the both-grouped CQR gain is only 1.12%. On the confirmation families, the same grouping yields a 3.27% gain at $h = 24$, but coverage remains 88.56%, compared with 87.87% for shared ACI. Grouping is thus an informative control, not a demonstrated solution to conditional reliability.

Appendix L. Detailed score constructions and original comparisons

Appendix L.1. Scores and inversions

Let $b_t = u_t - \ell_t$. We construct an outcome scale v_t from a causal exponentially weighted mean absolute error, updated only when a target is revealed, with decay 0.05. Its initialization uses the first available forecast band and point prediction, not an unseen outcome. The scale is bounded below by 10^{-12} for numerical arithmetic. Define $\tilde{b}_t = \max(b_t, 10^{-3}v_t)$. The four primary scores are

$$s_t^{\mathrm{R}} = |y_t - m_t|, \qquad s_t^{\mathrm{E}} = |y_t - m_t|/v_t, \tag{L.1}$$

$$s_t^{\mathrm{N}} = |y_t - m_t|/\tilde{b}_t, \qquad s_t^{\mathrm{Q}} = \max(\ell_t - y_t, y_t - u_t). \tag{L.2}$$

They yield intervals $[m_t - q_t, m_t + q_t]$, $[m_t - v_t q_t, m_t + v_t q_t]$, $[m_t - \tilde{b}_t q_t, m_t + \tilde{b}_t q_t]$ and $[\ell_t - q_t, u_t + q_t]$, respectively. The native-scale control asks whether a native width helps without using native endpoints directly; the past-error-scale control asks whether a simple outcome-based scale suffices. CQR thresholds may be negative, allowing a band to shrink.

If inversion produces reversed endpoints, we record the event and emit a singleton at their center. This is a practical interval repair rather than the exact empty set obtained from signed-score inversion. Coverage-driven updates use the emitted interval. Infinite intervals, if produced, are counted and retain infinite interval scores; they are not capped or dropped. The primary CQR–ACI test intervals are singletons at an average rate of 0.013%, and no infinite intervals occur in the reported primary comparison.

We also examine normalized CQR, with denominator $\max(b_t + \lambda v_t, 10^{-3}v_t)$ for $\lambda \in \{0, 0.01, 0.1, 0.5, 1\}$. A prefix selector chooses the lowest validation interval score among these five recipes and the four primary scores under ACI. This is an existing model-selection principle used as a diagnostic, not a proposed novel algorithm. A separate midpoint control replaces m_t by $(\ell_t + u_t)/2$ while retaining the residual or native-scale score.

Table L.16: Original one-step main-panel methods. Gain uses default residual–ACI; Gap and Local denote absolute overall and 150-origin coverage errors in percentage points.

Score / rule	Gain (%)	Cov. (%)	Gap (pp)	Local (pp)
residual/aci	0.00	89.94	0.07	1.66
error-scale/aci	2.17	90.00	0.03	1.06
native-scale/aci	11.50	89.98	0.05	1.42
cqr/aci	13.57	89.98	0.05	1.45
residual/faci	8.57	89.19	0.81	1.86
cqr/faci	14.87	89.49	0.52	1.71
residual/saocp	3.32	92.54	2.64	2.92
cqr/saocp	12.40	90.73	1.49	2.07
cqr/pi-log	13.88	90.00	0.06	1.51
cqr/sfogd	13.28	90.00	0.03	0.98
prefix-selected/aci	13.72	89.98	0.05	1.46
residual/aci/tuned	5.23	89.74	0.28	1.65
cqr/aci/tuned	14.18	89.92	0.15	1.74

Table L.17: Earlier CQR rate sensitivity at $h = 24$. Development and traffic/exchange-rate panels each have 16 pairs. S/I denote shared/issued. Selection uses validation only.

Panel	Grid	Gain	Wins	Cov. S	Cov. I	Local S	Local I
Development	original	6.54	15/16	88.18	89.56	5.33	5.55
Development	scaled	0.73	13/16	89.54	89.35	5.62	5.60
Development	wide	0.59	13/16	89.45	89.32	5.66	5.61
Traffic/FX	original	8.43	16/16	87.87	89.43	5.79	6.12
Traffic/FX	scaled	1.27	13/16	89.00	89.03	6.14	6.25
Traffic/FX	wide	0.99	13/16	88.78	89.01	6.21	6.22

Table L.18: Original one-step PV decomposition. Contributions use each pair's overall residual denominator and add to its overall gain; group gains use group denominators.

Settings	Group	Share	Gain	Contribution	Cov. R	Cov. CQR
default	all	100.00	35.25	35.25	89.88	89.96
default	positive	38.84	21.32	15.61	74.38	84.96
default	zero	61.16	73.37	19.64	99.72	93.14
tuned	all	100.00	20.51	20.51	89.34	89.72
tuned	positive	38.84	12.77	10.43	80.17	86.32
tuned	zero	61.16	54.89	10.08	95.16	91.88

Appendix M. Detailed new-source protocol and reference comparison

Appendix M.1. Protocol-locked confirmation on two new sources

After the preceding sensitivities, we fixed an additional protocol before numeric data acquisition or calibration inspection. It selects MPI-BGC Jena roof temperature and pressure for the first 3,600 hours of 2026, and PM2.5 at the Aotizhongxin and Dingling stations for the first 3,600 hours from 1 March 2013 in the UCI Beijing archive [25, 26]. These are four series from two sources, evaluated with the same two cached Chronos checkpoints: eight dependent model–series pairs. Both models emit all 24 horizons at 2,976 consecutive origins. The main test segment has 1,776 origins before excluding missing targets. Context length, batch size, dtype, checkpoint-file hashes, timestamps and generation code are recorded for every archive.

Missing input observations are forward filled causally; missing targets are excluded from score memory, controller increments and evaluation. Windows contain the latest W observed, revealed scores. Issued-state phases continue without an increment when their outcome is missing. Local coverage uses 150 calendar-origin windows with their observed denominators. The weather forecast targets are complete; the one missing initial weather hour precedes all forecast contexts. The new-source protocol is locally timestamped and hashed, not externally preregistered. Public historical air-quality data may have been used in neural pretraining; new-to-calibration does not imply unseen-to-model.

For both scores and states, nested grids comprise the original rates at $W = 450$; the wide grid described below; and that grid augmented with zero and four smaller rates (10^{-6}, 5×10^{-6}, 10^{-5} and 5×10^{-5}). The final stage crosses these expanded rates with $W \in \{200, 450, 900\}$. Validation selects the minimum interval score, breaking ties toward smaller rate and then window. Zero-rate trajectories provide both fixed-window and selected-window rolling controls. Early and late validation intervals, $[600+h-1, 900-h+1)$ and $[900, 1200-h+1)$, probe selector stability without changing the primary full-validation selector or test segment.

The wide grid combines rates 0.0001, 0.0002, 0.0005, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05 and 0.1 with the three original rates divided by h.

The predefined complete-weather subset also receives full AcMCP score-casting: the pinned author's MA($h-1$) plus cross-horizon regression implementation, at all horizons 1–24, with its supported CSS fitting option. We identify it as AcMCP-CSS, not the default CSS-ML configuration. Rolling

history is 600, $C_{\text{sat}} = 0.5$, and K_I uses only the first 600 revealed target-clock error rows. The rate grid is $\{0.001, 0.005, 0.01, 0.05, 0.1\}$ and these rates divided by h; the same validation block selects it. Fitted scorecasts are independent of this rate, so they are cached once per archive. A cached interval replay must agree with every original R trajectory at rate 0.01 before evaluating the other rates. A matched no-scorecast PID control uses the identical rate grid. Both are signed-error complete configurations; their comparison to native CQR does not isolate controller quality.

Appendix M.2. PV support and asymmetric-residual controls

On all 32 original one-step PV pairs, we additionally compare the established signed-error construction $[m_t + Q_{a_t/2}(Y - m), m_t + Q_{1-a_t/2}(Y - m)]$ with symmetric residuals and CQR. It shares the total-miscoverage ACI state, timing and window. Each construction uses the original nine rate/window settings; fixed defaults and a separately selected zero-rate rolling control are retained. This is exploratory analysis of previously inspected data, distinct from the new-source confirmation.

All three constructions are evaluated raw and after the same known-support projection $[L, U] \mapsto [\max(0, L), \max(0, U)]$. Projected and raw configurations are separately selected on validation score, while each candidate keeps its raw feedback trajectory: the projection isolates interval geometry and is not a new feedback algorithm. For a nonnegative target and finite ordered endpoints, projection cannot worsen interval score or lose a covered target. In particular, when $L < 0 \leq U$, score improvement is exactly $-L$, with unchanged coverage. When $U < 0$, both endpoints become zero and previously uncovered zeros can become covered. We therefore report coverage changes as well as width and miss-distance penalties.

In addition to actual zero/positive targets, an observable grouping compares each point forecast with the median of the first 600 point forecasts. This threshold is fixed before validation; the high-forecast group is known at issue time and is not selected using the future outcome. It is a diagnostic control, not a conditional-validity guarantee.

Appendix N. Post-review temporal and history controls

These analyses revisit inspected data. For positive-rate selectors, each state minimizes the original full-validation score over the original stored positive rates, then selects its window when allowed. The smallest rate is 10^{-6}. No candidate is added after test inspection. The air-quality CQR gaps are 0.0017% at $W = 450$ and 0.0021% with selected memory before admitting

Table M.19: Original configured weather comparison using full validation. Ratios use window-selected Rolling CQR; this precedes the common-history sensitivity.

Horizon	Method	IS ratio	Coverage	Local
1	Rolling CQR	1.000	89.37	3.15
1	Issued CQR	0.996	89.44	1.43
1	AcMCP-CSS	1.024	89.23	2.03
1	PID, no scorecast	1.026	89.20	1.99
6	Rolling CQR	1.000	90.27	2.91
6	Issued CQR	1.002	90.44	2.91
6	AcMCP-CSS	1.087	88.65	3.03
6	PID, no scorecast	1.084	88.39	3.23
24	Rolling CQR	1.000	89.51	6.03
24	Issued CQR	1.000	89.43	6.01
24	AcMCP-CSS	1.147	87.54	7.49
24	PID, no scorecast	1.056	87.18	7.31

zero; these differences are practically negligible. The paired records retain all rates and score scales.

Each test block contains 592 forecast origins. Missing labels are excluded using the fixed observed mask. Local errors use 150-origin windows wholly within each block, so whole-test local error is not exactly the weighted average of block values. No configuration is reselected on a test block. Figure N.9 includes every source, model and block; early/late validation sensitivity is a different analysis.

The common-history sensitivity adds unmodified author AcMCP-CSS at $n_{\text{cal}} = 450$ to the saved 600 runs. Both use K_I from the first 600 revealed target-clock errors. Because history also changes startup, every method uses post-startup validation $[900 + h - 1, 1200 - h + 1)$ and unchanged testing $[1200, 2976)$. All evaluated scorecasts are finite. Each history selects its rate; joint selection considers both histories. CQR retains its zero/small-rate grid; PID uses AcMCP's rates with zero scorecasts. Each ratio uses its matched Rolling denominator. History 900 and CSS-ML are outside this sensitivity.

All 60 sampled history-450 scorecast refits reproduce stored values. Three emit an MA non-invertibility warning under CSS but return finite predictions. Their warnings and trajectories are retained. These results characterize the stated configuration, not default CSS-ML or every AcMCP setting.

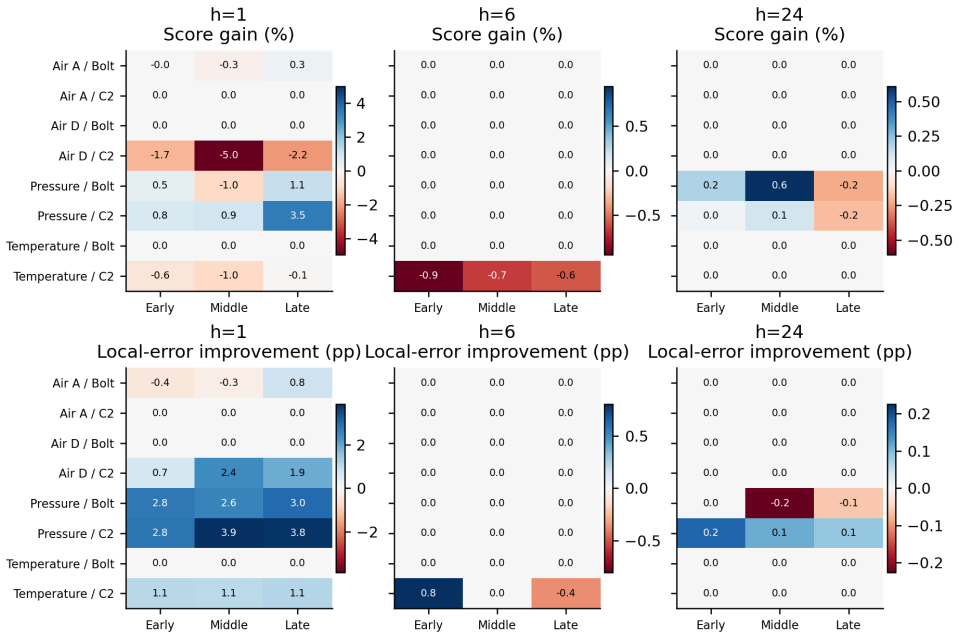


Figure N.9: All eight new-source pairs, three consecutive test blocks. Positive means issued CQR improves on matched selected-window Rolling CQR. Color scales differ between panels; values are descriptive, without independent-replication error bars.

Table N.20: Complete history sensitivity on all four weather pairs. IS ratios are averaged after pairing. Coverage is percent and Local is percentage points.

Horizon	History	Method	IS ratio	Coverage	Local
1	450	Rolling-CQR	1.000	89.64	3.32
1	450	CQR-shared	0.993	89.68	1.63
1	450	CQR-issued	0.993	89.68	1.63
1	450	AcMCP-CSS	1.023	89.25	2.08
1	450	PID-no-scorecast	1.024	89.32	1.86
1	600	Rolling-CQR	1.000	89.64	3.44
1	600	CQR-shared	0.995	89.60	1.71
1	600	CQR-issued	0.995	89.60	1.71
1	600	AcMCP-CSS	1.023	89.15	2.09
1	600	PID-no-scorecast	1.024	89.32	1.86
1	450/600	Rolling-CQR	1.000	89.58	3.40
1	450/600	CQR-shared	0.994	89.58	1.72
1	450/600	CQR-issued	0.994	89.58	1.72
1	450/600	AcMCP-CSS	1.023	89.33	1.98
1	450/600	PID-no-scorecast	1.024	89.32	1.87
6	450	Rolling-CQR	1.000	89.74	2.91
6	450	CQR-shared	1.018	89.96	2.45
6	450	CQR-issued	1.004	89.85	2.97
6	450	AcMCP-CSS	1.087	88.68	3.04
6	450	PID-no-scorecast	1.087	88.84	2.97
6	600	Rolling-CQR	1.000	89.95	2.83
6	600	CQR-shared	1.014	90.15	2.58
6	600	CQR-issued	1.008	90.06	2.68
6	600	AcMCP-CSS	1.085	88.67	3.09
6	600	PID-no-scorecast	1.091	88.94	2.97
6	450/600	Rolling-CQR	1.000	89.96	2.86
6	450/600	CQR-shared	1.029	90.09	2.24
6	450/600	CQR-issued	1.004	90.20	2.79
6	450/600	AcMCP-CSS	1.085	88.67	3.09
6	450/600	PID-no-scorecast	1.091	88.94	2.97
24	450	Rolling-CQR	1.000	89.32	6.33
24	450	CQR-shared	1.136	83.22	8.96
24	450	CQR-issued	1.000	89.27	6.34
24	450	AcMCP-CSS	1.138	89.18	6.45
24	450	PID-no-scorecast	1.061	88.20	6.49
24	600	Rolling-CQR	1.000	89.44	6.75
24	600	CQR-shared	1.039	87.43	7.42
24	600	CQR-issued	1.000	89.43	6.74
24	600	AcMCP-CSS	1.172	89.27	6.27
24	600	PID-no-scorecast	1.067	88.37	6.43
24	450/600	Rolling-CQR	1.000	89.44	6.75
24	450/600	CQR-shared	1.039	87.43	7.42
24	450/600	CQR-issued	1.000	89.43	6.74
24	450/600	AcMCP-CSS	1.164	89.18	6.48
24	450/600	PID-no-scorecast	1.065	88.60	6.24

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